# Example

- Does taking iron during pregnancy affect hemoglobin at delivery?
  - Consider a small study, with 200 participants, 100 receiving iron, 100 receiving placebo – we'll do this example in SAS in a later Topic
- Explore and check your data – get to know it
- Summary stats, graphs

| If Someone is Accused of a Crime… | Hypothesis Testing |
|---|---|
| Did they commit the crime? | Does hemoglobin at delivery depend on whether a person takes iron supplements during pregnancy? |
| Starting assumption: nothing happened: they are innocent | Null hypothesis: nothing is happening: iron supplementation during pregnancy does not affect hemoglobin at delivery |
| Alternatively, they are guilty | Alternative hypothesis: iron does affect hemoglobin at delivery |
| Gather evidence | Gather data → $P\left(\text{observing data as extreme as/more than actually observed} \mid H_0 \text{ true}\right)$ <br> Conditional Probability |
| How possible is it that they are innocent and all the evidence happened only by chance? | P-value! The probability that iron does not affect hemoglobin at delivery but you still see data as extreme as yours or even more extreme <br> Ex): p-value = 0.02, Reject $H_0$ (relationship ↑) <br> (2% chance you'd get result this extreme or more just by random chance). ← test-Statistic <br> ∘ Evidence against $H_0$. |
| Judgment:<br>• "Guilty": the evidence is very unlikely to happen only by chance<br>• "Not Guilty": the evidence is not strong enough – maybe it's chance | Decision:<br>• "Reject null hypothesis": $p < \alpha$ (usually $\alpha = 0.05$, so $p < 0.05$)<br>• "Do not reject null hypothesis": $p \geq \alpha$ (usually $\alpha = 0.05$, so $p \geq 0.05$) <br> ⇓ $0.02 < 0.05$ <br> ↓ Probability to be under $H_0$ region |
| Does "Not Guilty" mean innocence? | Can we "accept" the null hypothesis? |

# Hypothesis Testing

1. State the scientific question to be answered

**Does taking iron during pregnancy affect hemoglobin at delivery?**

2. Define the parameters of interest

**$\mu_1$ = mean hemoglobin (in g/dL) at delivery among women (of our population) who take iron during pregnancy**

**$\mu_2$ = mean hemoglobin (in g/dL) at delivery among women (of our population) who take a placebo/do not take iron during pregnancy**

**(parameter vs. estimate)**

# Hypothesis Testing

3. State the null hypothesis $H_0$ both mathematically (in terms of parameters) and in words

   - Usually an equality

   $H_0$: $\mu_1 = \mu_2$

   **Women in this population who take iron during pregnancy have the same mean hemoglobin at delivery as women in this population who do not take iron during pregnancy**

4. Decide whether the test is directional or not, and state the alternative hypothesis ($H_A$ or $H_1$) both mathematically and in words

   $H_A$: $\mu_1 \neq \mu_2$ **(not directional)**

   **In this population, the mean hemoglobin at delivery is different among women who take iron during pregnancy compared with women who do not take iron during pregnancy**

# Hypothesis Testing

5. Name the type of statistical test to be used and <u>the distribution the test statistic follows under $H_0$</u>
   - Determine appropriate test statistic based on null hypothesis
   - Include the degrees of freedom (or other parameters) for that distribution if appropriate

**t test**

**t statistic**

**$t \mid H_0 \sim t_{n-2}$**

**n=200, so df=198**

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

# Hypothesis Testing

6. State the significance level $\alpha$ to be used and the corresponding critical value of the test statistic. Define the rejection region.

- The critical value and rejection region depend on $\alpha$, the df, the distribution from #5, and whether $H_A$ is directional

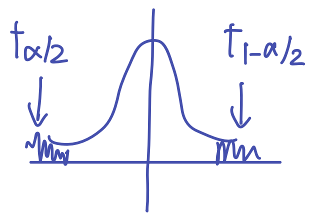
**$\alpha$ = 0.05**

**Critical values: $t_{\alpha/2,df=n-2}$ and $t_{1-(\alpha/2),df=n-2}$** ← Draw a picture!

**Critical values: $t_{0.025,df=198}$ and $t_{0.975,df=198}$**

**Critical values: -1.972 and 1.972**

**Reject $H_0$ if $t < -1.972$ or $t > 1.972$**



# Hypothesis Testing

7. Calculate the test statistic from the data
8. Compare the test statistic to the rejection region, or give the p-value and compare it to $\alpha$.
   - P-value < or > $\alpha$?

**$t = 3.17$**

**$3.17 > 1.972$**

**$p = 0.0018 < 0.05$**

# Hypothesis Testing

9. Make a decision about the null hypothesis:
   a) If the test statistic is in the rejection region, we "reject $H_0$"
   b) If not, we "do not reject $H_0$"

   We DO NOT use the word "accept" anywhere in this step.

   If using p-value, reject $H_0$ when $p < \alpha$.

   **We reject $H_0$**

# Hypothesis Testing

10. Form a scientific conclusion based on your decision.

If 9a), then start with "This study provides evidence…"
If 9b), then start with "This study does not provide evidence…"

followed by "at the ___ significance level that"

followed by the verbal statement of $H_A$.

**This study provides evidence at the 0.05 significance level that in this population, the mean hemoglobin at delivery is different among women who take iron during pregnancy compared with women who do not take iron during pregnancy**